

Teaching Exam

**Subject –
Computer
Science**

Operating System.



Lecture No –1

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Topics to be Covered

Software

Topic

Operating System

System Software



Operating System

Introduction to Python //

Data handling using Pandas

Emerging Trends //

Society, Law, Ethics.

Social trends

Operating System //

DSA //

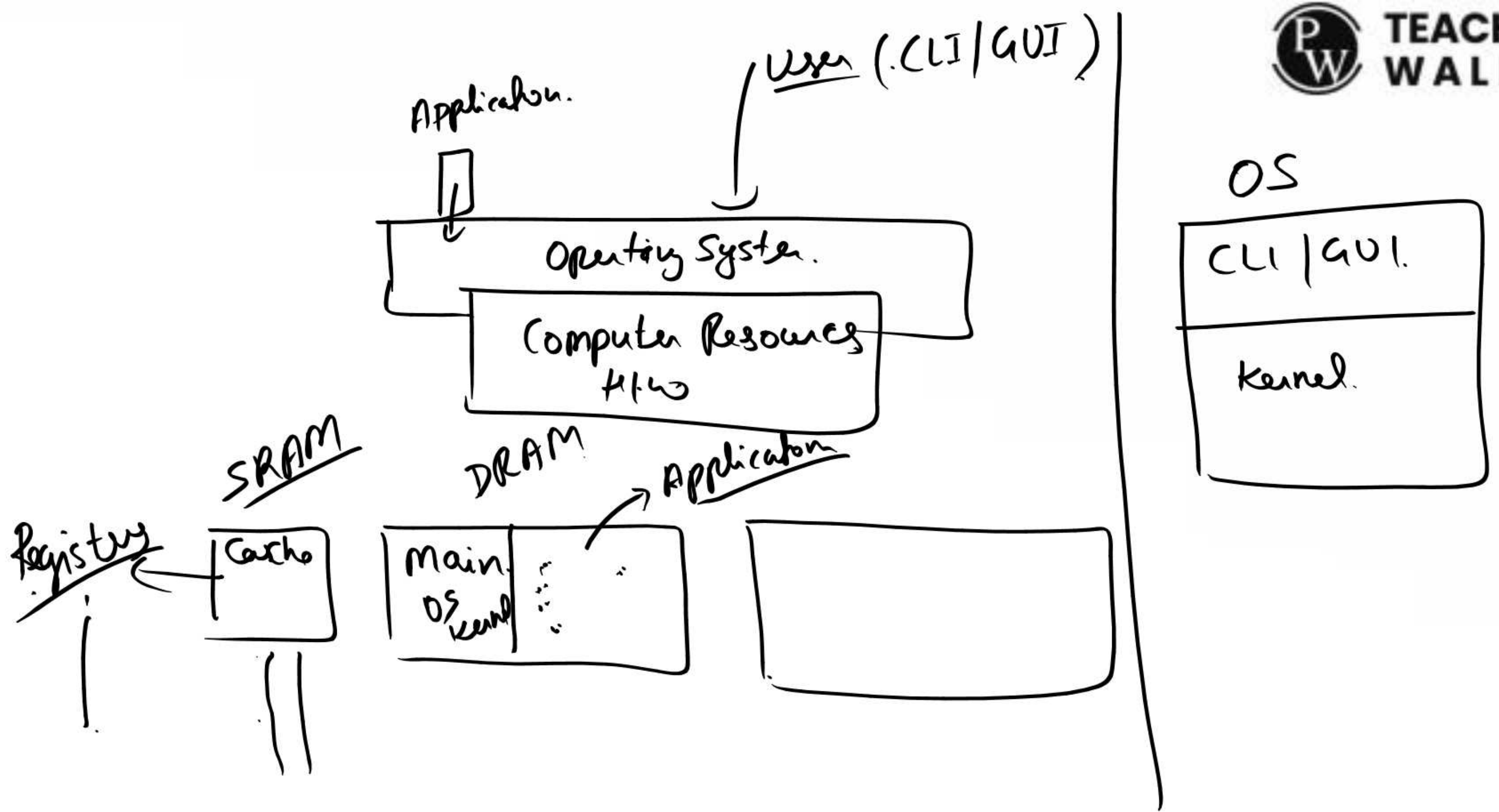
Computational thinking & Programming

||

= 2

Web based Application

Operating Systems and Windows



What is an Operating System?

- The **most important** program that runs on your computer. It **manages** all other programs on the machine.
- Every PC **has to have one** to run other applications or programs. It's the first thing **"loaded"**.



Operating System

- It performs basic tasks, such as:
 - **Recognizing input** from the keyboard or mouse,
 - **Sending output** to the monitor,



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Operating System

- **Keeping track of files and directories on the disk, and**
- **Controlling peripheral devices such as disk drives and printers.**



Is There More Than One Type of OS?

- Generally, there are four types, based on the **type of computer they control** and the **sort of applications they support**.

1. Single-user, single task

This type manages the computer so that one user can effectively do one thing at a

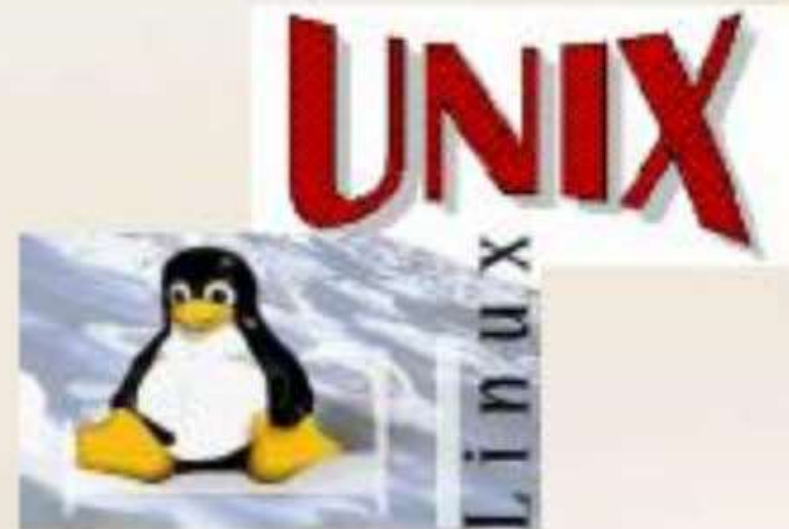


Types of Operating Systems

2. Multi-user, multi-task

Allows **two or more users to run programs at the same time**. Some operating systems permit hundreds or even thousands of concurrent users.

Mainframe



Types of Operating Systems

3. Real Time Operating Systems

RTOS are used to control machinery, scientific instruments, and industrial systems.



There is typically very little user-interface capability.

Resources are managed so that a *particular operation executes precisely the same every time*



Types of Operating Systems

4. Single-user, Multi-tasking

*This is the type of operating system **most desktops and laptops use today.***

*Microsoft's Windows and Apple's MacOS are both examples of operating systems that will let a **single user have several programs in operation at the same time.***



OS's Manage Applications

- Operating systems provide a software platform on top of which other “application” programs can run.
- The application programs must be written to run on a particular operating system.
- So, your choice of operating system determines what application software you can run



Operating System Functions

- Besides managing hardware and software resources on the system, the OS must **manage resources and memory.**
- **There are two broad tasks to be accomplished.**

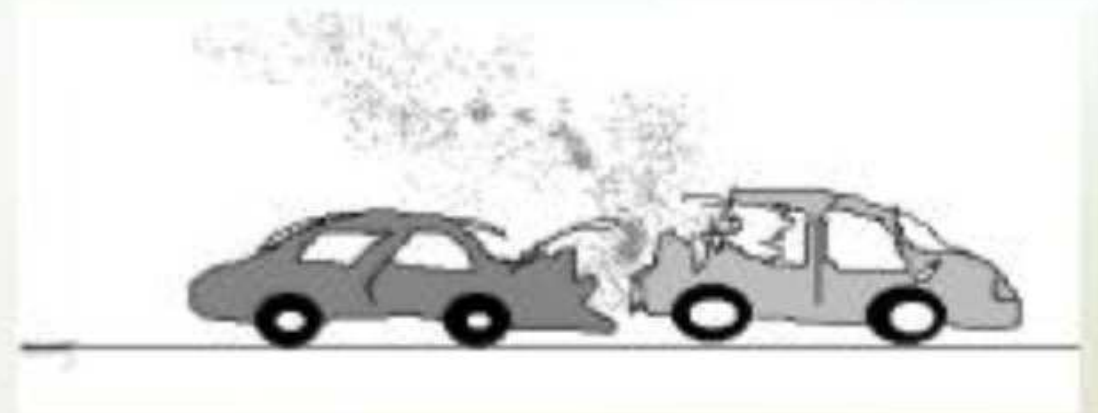


OS - Memory Storage and Management

1. Each process **must have enough memory** in which to execute, and

It can **neither run into** the memory space of another process,

Nor be run into by another process



OS - Memory Storage and Management

1. The different types of memory in the system **must be used properly** so that each process can run most effectively.



Cache Memory

Cache Memory.

SRAM. (Flip flops)

DRAM
RAM (capacitors)

- **Cache** - A section of a computer's memory which temporarily retains recently accessed data in order to speed up repeated access to the same data.

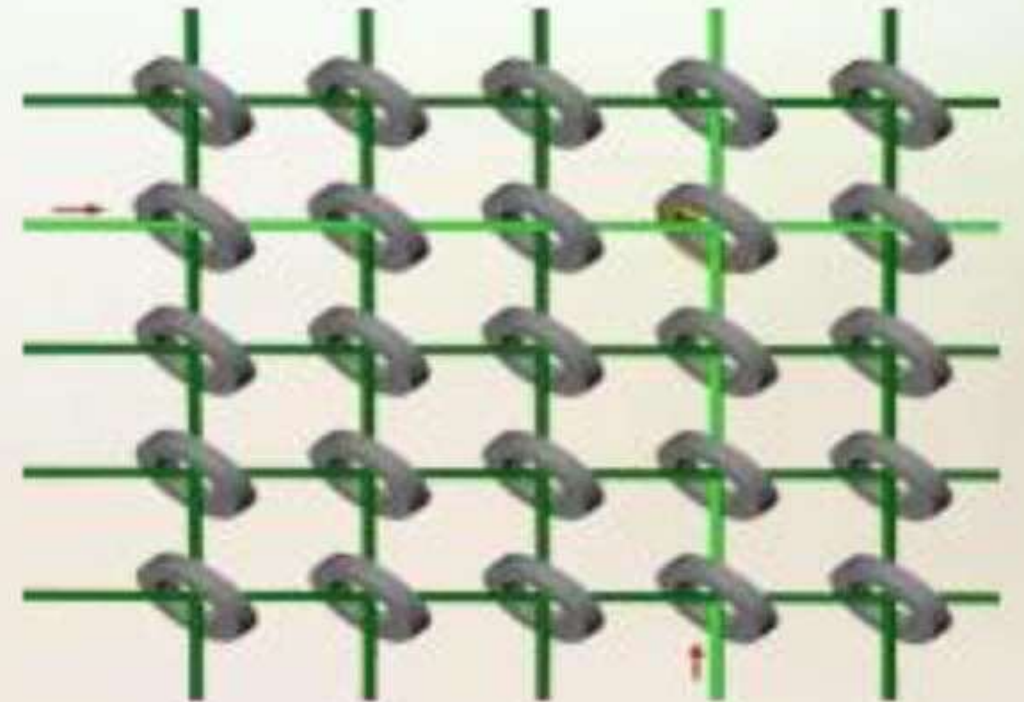
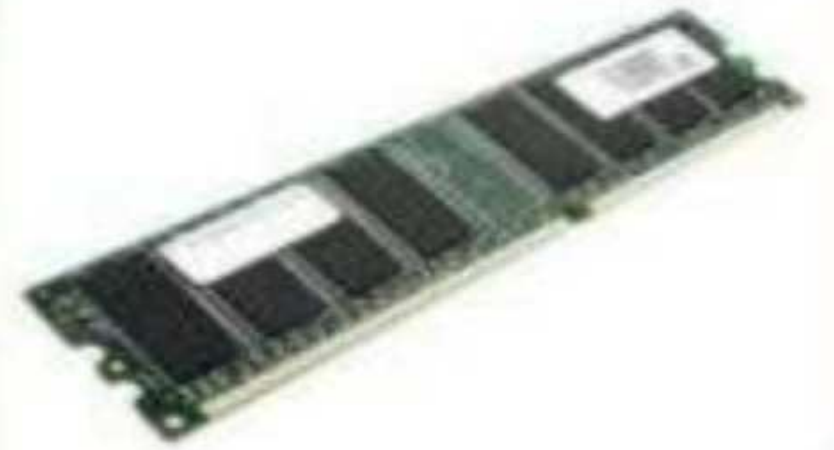
- It provides rapid access without having to wait for systems to load.



RAM Memory

- **Random access memory (RAM)** is the best known form of computer memory.

- RAM is considered "random access" because you can **access any memory cell directly** if you know the row and column that intersect at that cell.



RAM Memory

- The **more** RAM your computer has, the **faster** programs can function. The two main types are called DRAM and SRAM. SRAM is faster than DRAM, but, more expensive.

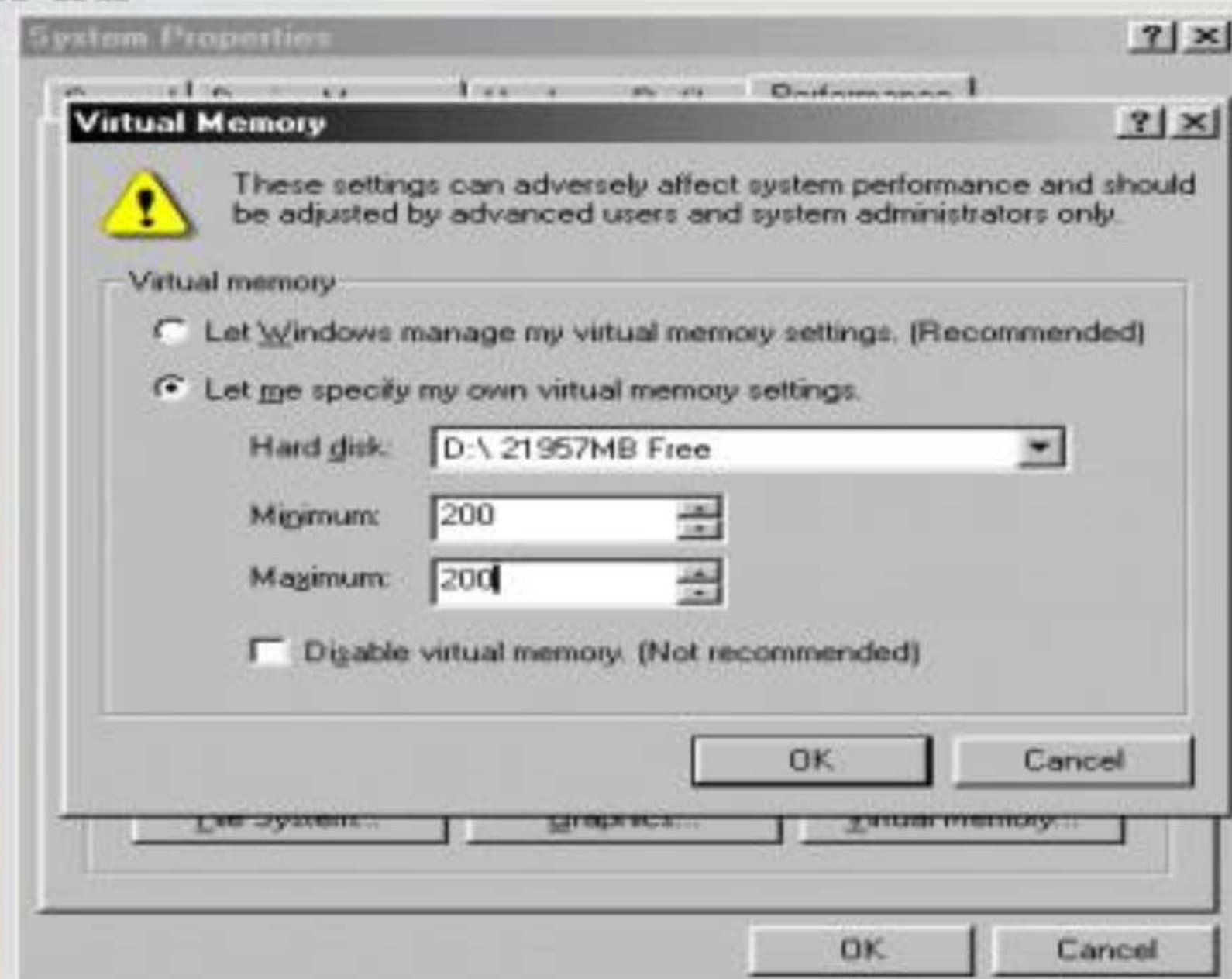
Remember, that if the power is turned off, then all data left in RAM, that has not been saved to the hard drive, is lost.

Virtual Memory

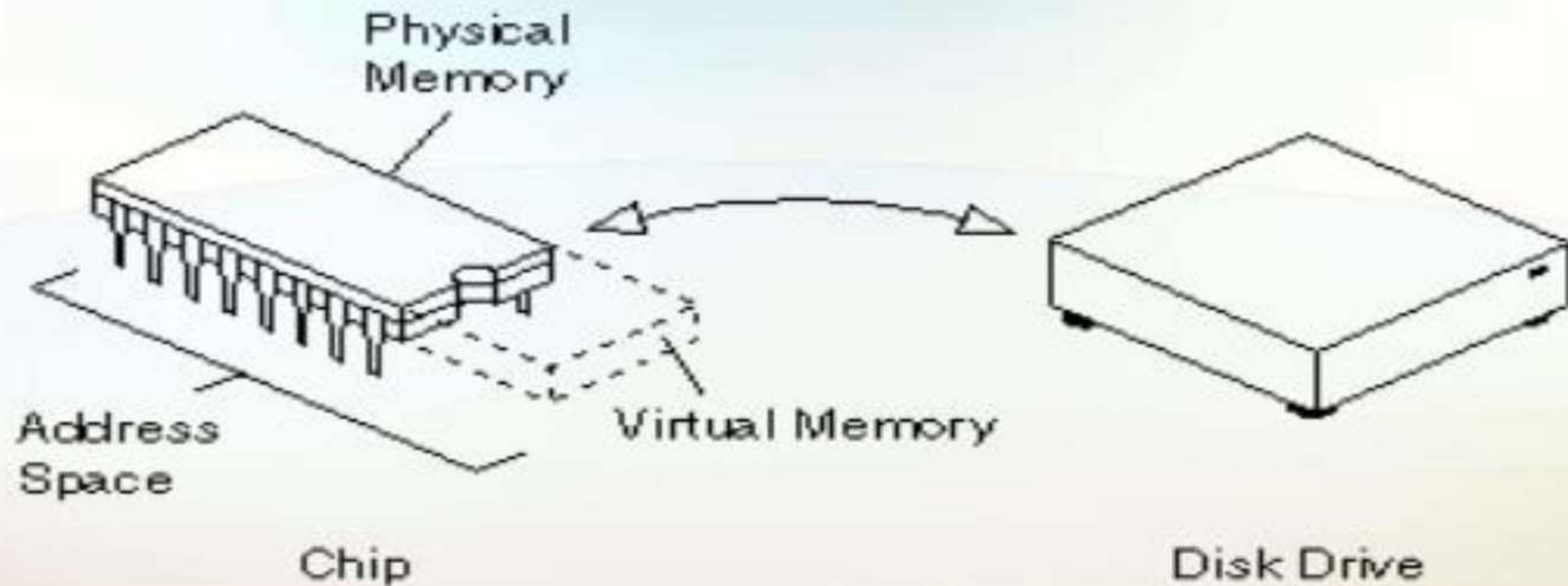
Handwritten notes:
4GB
1.5GB OS ker.
7.5GB

- **Virtual Memory** – a method of using **hard disk space** to provide **extra** memory. It simulates additional RAM

- In Windows, the amount of virtual memory available, equals the amount of **free RAM** plus the amount of **disk space** allocated to the **swap file**.



Virtual Memory – Swap File



A swap file is an area of your hard disk that is set aside for virtual memory. Swap files can be either temporary or permanent.

Okay – So Now What?





OS - Wake up call

- When you **turn on the power** to a PC, the first program that **runs** is a set of **instructions** kept in the computer's read-only memory (ROM).



OS - Wake up Call



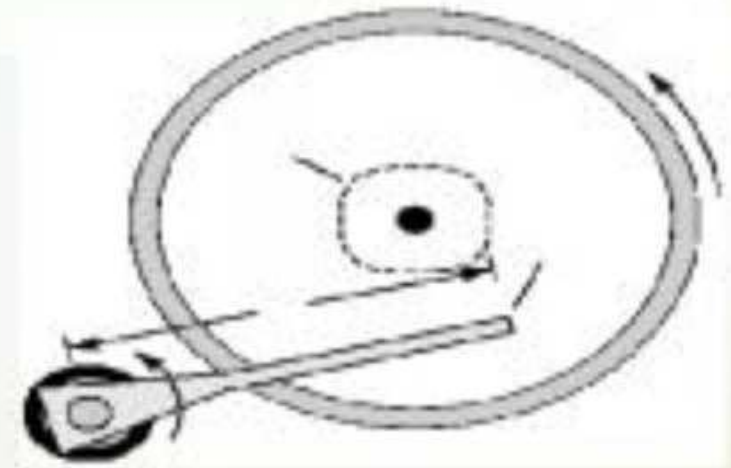
- It checks to make sure everything is **functioning properly**.

- It **checks** the CPU, memory, and basic input-output systems (BIOS) **for errors**.



OS – Wake up Call

- Once successful, the software will begin to **activate** the computer's **disk drives**.



- It then **finds** the first piece of the operating system: the **bootstrap loader**.

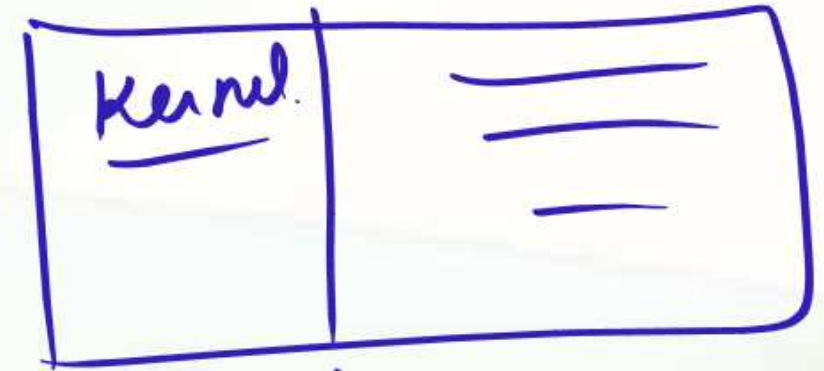


OS - Booting the PC

- The **bootstrap loader** is a small program that has a single function: It **loads the operating system** into memory and allows it to begin operation.

OS - Booting the PC

- The bootstrap loader sets up the small driver programs that interface with and control the various hardware.

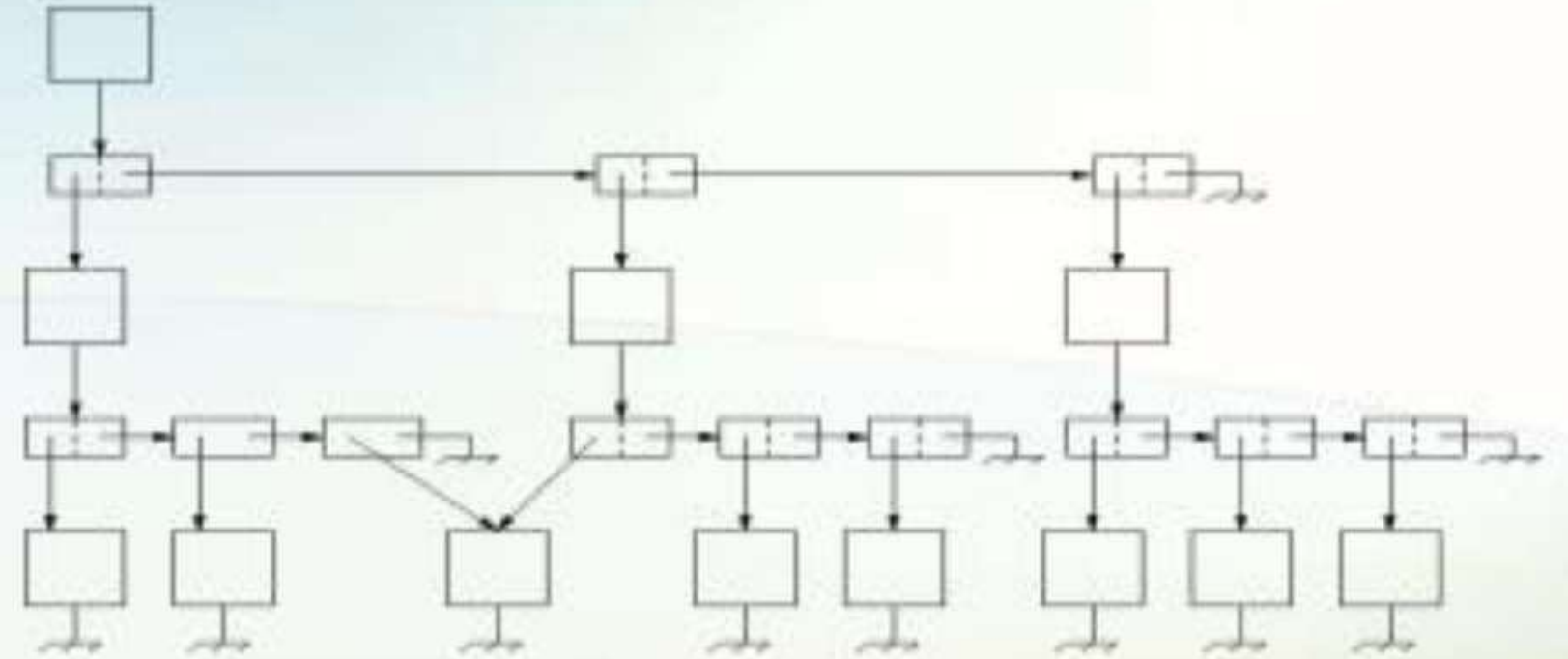


- It sets up the divisions of
 - memory
 - user information, and
 - applications.

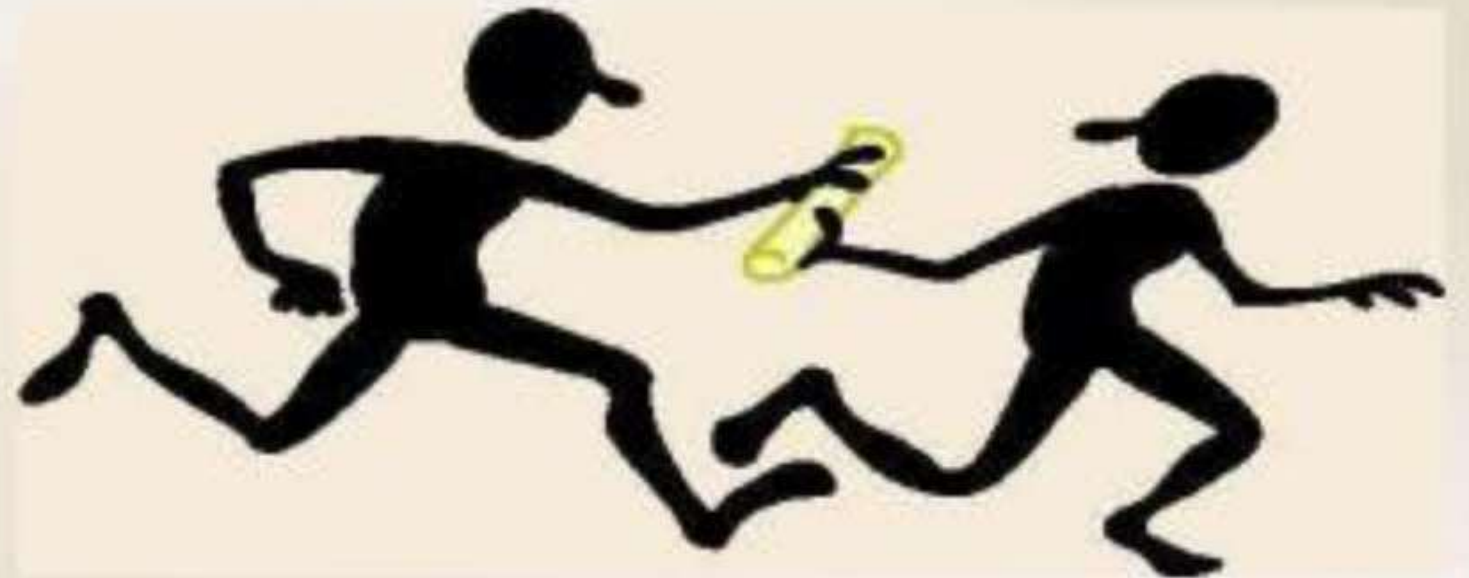


OS - Booting the PC

- It establishes the **data structures** needed to **communicate** within and between the subsystems and applications of the computer.



- Then it **turns control** of the computer over to the operating system.





2 mins Summary



TEACHING
WALLAH

Topic

Course (coding)

OS → bank

THANK YOU